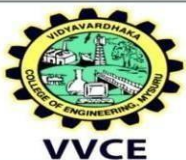


| COURSE CONTENT AND PLANNING                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| SEMESTER-V                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |
| Course Name: Numerical Methods and its Applications                                                                                                                                                                                                                                                                                                                                                                                                                                              | Course Code: BMANM561 |
| No. of Lecture hours / week: 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CIE Marks: 50         |
| No. of Tutorial/Practical hours / week: 0                                                                                                                                                                                                                                                                                                                                                                                                                                                        | SEE Marks: 50         |
| Total No. of Lecture Hours + Tutorial/Practical hours: 36                                                                                                                                                                                                                                                                                                                                                                                                                                        | SEE Duration: 3 Hrs   |
| L: T: P: 3:0:0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Credits: 3            |
| <b>COURSE PREREQUISITES:</b> Algebra of Matrices, Elementary Numerical Techniques,                                                                                                                                                                                                                                                                                                                                                                                                               |                       |
| <b>COURSE OVERVIEW:</b> The objective is to enable the students to apply the knowledge of numerical techniques in various fields of Engineering with applications                                                                                                                                                                                                                                                                                                                                |                       |
| <b><u>COURSE LEARNING OBJECTIVES</u></b><br>The objective is to enable the students to apply the knowledge of advance numerical Mathematics <ul style="list-style-type: none"> <li>• To find approximate solution of polynomial and simultaneous equation</li> <li>• To evaluate numerical values of differentiation and integration</li> <li>• To evaluate approximate value of eigenvalue and eigenvector of symmetric matrix</li> <li>• To solve differential equation numerically</li> </ul> |                       |
| COURSE CONTENT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                       |
| <b>Module I: Roots of Polynomial</b><br>Roots of Polynomials: Bairstow's method, Graeffe's Roots Squaring Method.<br><b>SLT:</b> Approximations and round off errors: Significant figures, accuracy and precision, error definitions, roundoff errors and truncation errors.                                                                                                                                                                                                                     |                       |
| <b>Module II: Solution of Linear System - Direct and Iterative Methods</b><br>Matrix Partition method, Positive Definite matrix and LU- Decomposition, Cholesky decomposition<br><b>SLT:</b> Least Square method for Inconsistent systems.                                                                                                                                                                                                                                                       |                       |
| <b>Module III: Numerical Differentiation and Numerical Integration</b><br>Divided difference interpolation, Numerical Differentiation (Newton Forward, backward and divided difference). Romberg integration and Double integral using Simpson's (1/3) rd rule.<br><b>SLT:</b> Maxima and Minima using interpolation                                                                                                                                                                             |                       |
| <b>Module IV: Eigen Values and Eigen Vectors of Symmetric matrices</b><br>Eigen Values and Eigen Vectors of symmetric matrices using Jacobi and Givens method for symmetric matrices, Householder's method for symmetric matrices.<br><b>SLT:</b> Rutishauser method for arbitrary matrices.                                                                                                                                                                                                     |                       |
| <b>Module V: Numerical solution for IVP and BVP</b><br><b>IVP:</b> Picard's method (first order)<br><b>BVP:</b> Successive Over Relaxation method and Alternating Direction Implicit method.<br><b>SLT:</b> Picard's method (second order)                                                                                                                                                                                                                                                       |                       |
| Text Books                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                       |
| 1. S.S.Sastry, Introductory Methods of Numerical Analysis, PHI, 2005.                                                                                                                                                                                                                                                                                                                                                                                                                            |                       |



2. Steven C. Chapra, Raymond P. Canale, Numerical Methods for Engineers, Tata McGraw Hill, 4<sup>th</sup> Ed., 2002.
3. M K Jain, S.R.K. Iyengar, R K Jain, Numerical Methods, for Scientific and Engg. Computation, New Age International, 2003.
4. Madhumangal Pal, Numerical Analysis for Scientists and Engineers, Narosa Publishing House, Latest Edition.

**Reference Books:**

1. Pervez Moin, Fundamentals of Engineering Numerical Analysis, Cambridge, 2010.
2. David C. Lay, Linear Algebra and its Applications, 3<sup>rd</sup> edition, Pearson Education, 2002.

**Course Outcomes:** At the end of this course, students will be able to

|                 |                                                                                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------|
| <b>C01</b>      | <b>Understand</b> the numerical methods for solving problems in Calculus and Linear Algebra <b>(PO-1)</b> . |
| <b>C02</b>      | <b>Apply</b> the concept numerical methods in Calculus and Linear Algebra <b>(PO-1)</b> .                   |
| <b>C03</b>      | <b>Apply</b> the concept numerical methods in Calculus and Linear Algebra <b>(PO-2)</b> .                   |
| <b>C04</b><br>* | Communicate the concepts of mathematical modeling through presentation.                                     |

**SEE - Course Assessment Plan**

| CO         | Marks Distribution |           |           |           |           | Total Marks | Weightage (%) |
|------------|--------------------|-----------|-----------|-----------|-----------|-------------|---------------|
|            | Module-1           | Module-2  | Module-3  | Module-4  | Module-5  |             |               |
| <b>C01</b> | 10                 | -         | -         | --        | --        | 10          | 10%           |
| <b>C02</b> | 10                 | 10        | 8         | 20        | 20        | 68          | 68%           |
| <b>C03</b> | -                  | 10        | 12        | -         | -         | 22          | 22%           |
|            | <b>20</b>          | <b>20</b> | <b>20</b> | <b>20</b> | <b>20</b> | <b>100</b>  | <b>100%</b>   |

**Intra-Module Choice for Answering of Questions**

|                                                        |                    |
|--------------------------------------------------------|--------------------|
| <b>Module - III</b>                                    | <b>Module - IV</b> |
| In other modules there will be no choice for answering |                    |